

Design and Analysis of Algorithm(Past papers)

PP-1:-

Q.6. Explain the following:

(4x5=20)

- 1) Define dynamic programming. What are features of dynamic programming?
- 2) Explain big-theta and Little-Omega. Write their set definitions and also draw figures to explain.
- 3) Draw recursion tree to analyze worst case of quick sort. Why is it worst?
- 4) Write invariant of quick-sort algorithm.

Q.7.

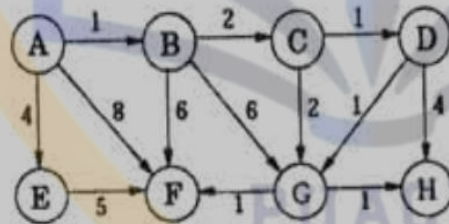
A) Component of a Graph: "In graph theory, the complement or inverse of a graph G is a graph H with vertices such that two vertices of H are adjacent if and only if they are not adjacent in G ."

Assume that graph with n vertices is represented by adjacency matrix $G[n][n]$. Give Pseudocode/Algorithm taking G , as input and giving as output the corresponding adjacency matrix of complement graph H . [10]

B) Write down steps to find strongly connected components from a digraph. [10]

Q.8.

Suppose Dijkstra's algorithm is run on the following graph, starting at vertex A. If you find a tie while selecting a vertex, then resolve the tie by selecting alphabetical order.



The table is storing following three pieces of information for each vertex 'V'.

visited = set to True if 'V' is visited, False otherwise

dist = Distance of the vertex 'V' from source vertex

prev = prior vertex connecting vertex 'V' to source vertex and improving its dist

A) Show the vertex visited on the top of each table and then show the updates made in table after visiting it. [15]

1) Initial state				2) After visiting _____				3) After visiting _____			
Vertices	visited	dist	prev	Vertices	visited	dist	prev	Vertices	visited	dist	prev
A	F	0	0	A				A			
B	F	∞	0	B				B			
C	F	∞	0	C				C			
D	F	∞	0	D				D			
E	F	∞	0	E				E			
F	F	∞	0	F				F			
G	F	∞	0	G				G			
H	F	∞	0	H				H			

4) After visiting _____

Vertices	visited	dist	prev
A			
B			
C			
D			
E			
F			
G			
H			

5) After visiting _____

Vertices	visited	dist	prev
A			
B			
C			
D			
E			
F			
G			
H			

6) After visiting _____

Vertices	visited	dist	prev
A			
B			
C			
D			
E			
F			
G			
H			

7) After visiting _____

Vertices	visited	dist	prev
A			
B			
C			
D			
E			
F			
G			
H			

8) After visiting _____

Vertices	visited	dist	prev
A			
B			
C			
D			
E			
F			
G			
H			

9) After visiting _____

Vertices	visited	dist	prev
A			
B			
C			
D			
E			
F			
G			
H			

B)

Redraw the graph, now showing only those edges and their weights making shortest paths to all vertices from source vertex.

[5]

PP-2:-

Q 2) Answer the following

[5+5+5 = 15]

- A) What is the difference between big-O and small-o notation?
- B) What will be the height of a max heap with 800 values?
- C) Write a function to find minimum of a binary search tree.

Q 3) What is the worst-case running time for the following:

[8]

Sorting Algorithm	Worst-case running time
Quick Sort	
Insertion Sort	
Heap Sort	
Merge Sort	

Q 4) Give recurrence relations of quick sort for best case and worst case. Give examples for both cases. (12)

Q 5) Show step by step construction of Min Heap from the following data. (7)

4 1 3 2 16 9 10 14 8 7

Q 6) Sort the following data using Merge Sort. Show all steps. (8)

23, 56, 85, 12, 0, 12, 14, 19, 17, 36, 25, 90, 66, 100, 9

Q 7-A) Show how we derive the recursion tree for $T(n) = 3T(n/4) + cn^2$. (10)

Q 7-B) What will be upper bound (big-O) in this case? (5)

Q 8-A) Consider the following graph:



Consider above graph and draw adjacency list. (5)

Q 8-B) Run Prim's algorithm step by step (show new graph at each stage) and find the minimum spanning tree. (10)

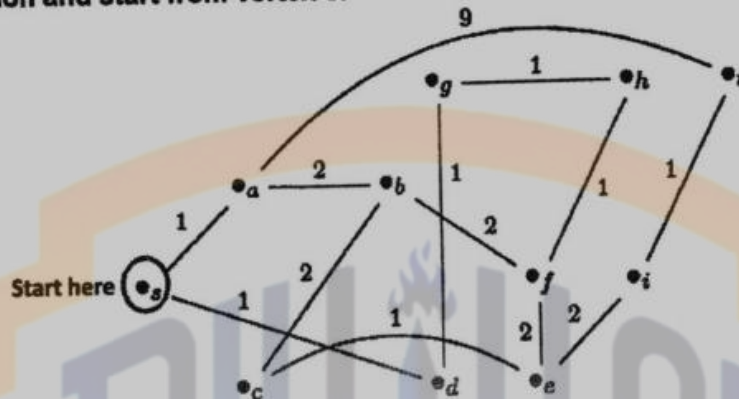
PP-3:-

Q.2 Suppose that you have a list of 1,000,000 grades (from 0% to 100%) on a nationwide test, and you want to put them in order (sort in any order). Design an algorithm that sorts this list in linear time ($O(n)$). Also, justify your choice of algorithm. [10+4]

Q.3 Fill in the following table with appropriate inputs, outputs and running times for each of the following algorithms. [8*2=16]

Algorithm	Input	Output	Time Complexity
DFS			
BFS			
Kruskal			
Prim's			
Dijkstra			
Floyd Warshall			
Knapsack Fractional			
Matrix Chain Multiplication			

Q.4 Draw the tree produced by running Prim's MST algorithm. Ignore edge orientation and start from vertex s . [20]



Q.5 Given below is a knapsack with maximum capacity $W=11$ and a set S consisting of 5 items with profits. How can we pack the knapsack to achieve maximal total value of packed items, when an item may either be Completely or Partially Selected? [15]

Item i	Weight w_i	Profit p_i
1	1	1
2	2	6
3	5	18
4	6	22
5	7	28

Q.6 The best-case running time for quick sort is $O(n \log n)$ whereas the worst-case running time is $O(n^2)$. Give recurrence relations of quick sort for both cases and discuss what types of input arrays result in best and worst case running times when quick sort is run on them, . [10+5]